



Teacher's Name _____

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Student Number

Knox Grammar School

2014

**Trial Higher School Certificate
Examination**

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time - 3 hours
- Write using blue or black pen only
- Board approved calculators may be used
- A table of standard integrals is provided at the back of this paper
- All necessary working should be shown in every question

Subject Teachers

Mr I Bradford
Mr D Sedgman

Setter

Mr M Vuletich

**This paper MUST NOT be removed from the
examination room**

Number of Students in Course: 34

Total Marks – 100

Section I 10 Marks

- Answer Questions 1 to 10
- Use the Multiple Choice Answer Sheet

Section II 90 Marks

- Answer Questions 11 to 16
- All questions are worth 15 marks
- Answer each question in a separate Writing Booklet.

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Section I

10 Marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10

1. Which of these ellipses has foci $(0, \pm 3)$?

A) $8x^2 + y^2 = 8$

B) $5x^2 + 4y^2 = 20$

C) $16x^2 + 25y^2 = 400$

D) $25x^2 + 16y^2 = 400$

2. Find $\int \cot x \, dx$

A) $-\operatorname{cosec}^2 x + c$

B) $-\ln(\operatorname{cosec} x) + c$

C) $\frac{1}{2} \cot^2 x + c$

D) $\ln(\sec x) + c$

3. The speed of a particle moving in a horizontal circle with radius 6 cm is $48\pi \text{ cm s}^{-1}$.

How many revolutions per second does this particle make?

A) 4

B) 240

C) 480

D) 480π

4. The polynomial equation $x^3 - 2x^2 + 3 = 0$ has roots α , β and γ .

What is the value of $\alpha^3 + \beta^3 + \gamma^3$?

A) -2

B) -1

C) -8

D) 8

5. Evaluate $\int_1^e \frac{dx}{x\sqrt{1+(\ln x)^2}}$.

A) $-\frac{\pi}{4} + \tan^{-1} e$

B) $\ln\left(\frac{e + \sqrt{e^2 + 1}}{1 + \sqrt{2}}\right)$

C) $\frac{\pi}{4}$

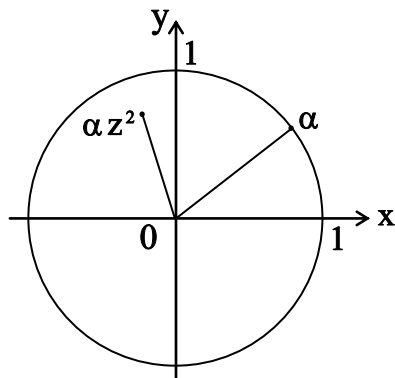
D) $\ln(1 + \sqrt{2})$

6. What is the simplest expression of

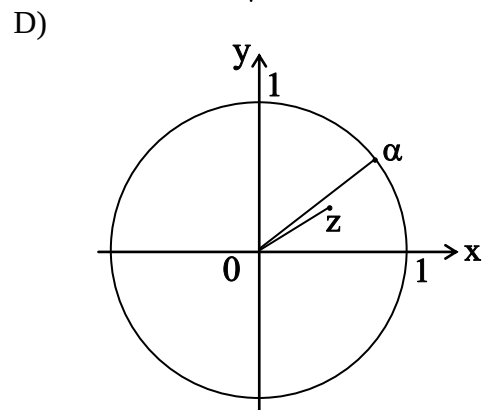
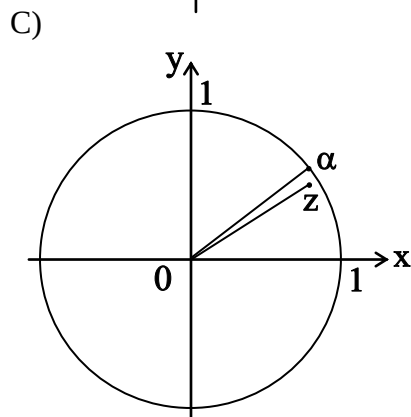
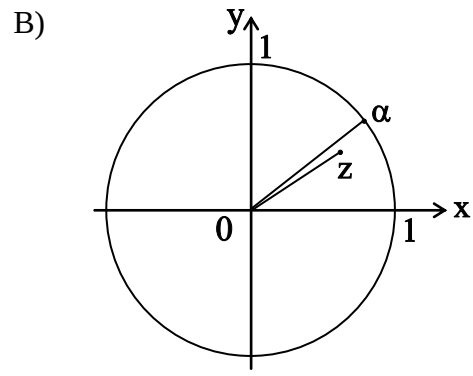
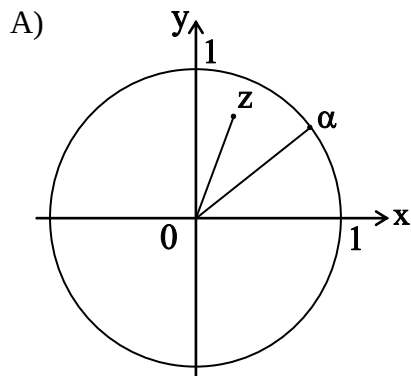
$$\text{Arg} \left(\frac{1}{1+i} \right) + \text{Arg} \left(\frac{1}{(1+i)^2} \right) + \text{Arg} \left(\frac{1}{(1+i)^3} \right) + \dots + \text{Arg} \left(\frac{1}{(1+i)^{20}} \right) ?$$

- A) $-\frac{\pi}{2}$ B) $-\frac{\pi}{4}$ C) $\frac{\pi}{4}$ D) $\frac{\pi}{2}$

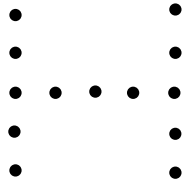
7. The Argand diagram below shows the complex numbers α and αz^2 .



Which of the following best represents the positions of z and α ?

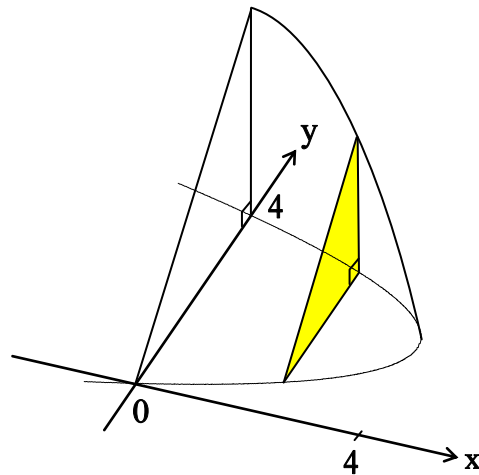


8. The diagram shows a shape made by 13 points.



How many triangles can be made with these points as vertices?

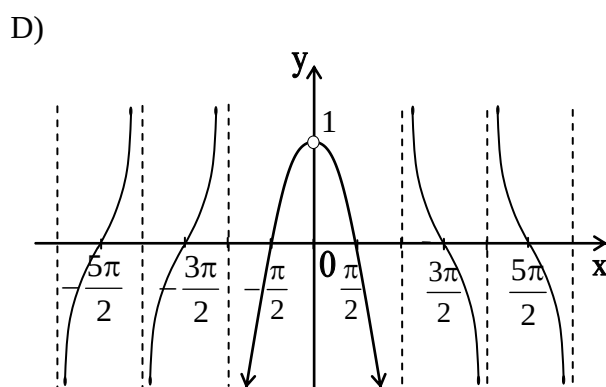
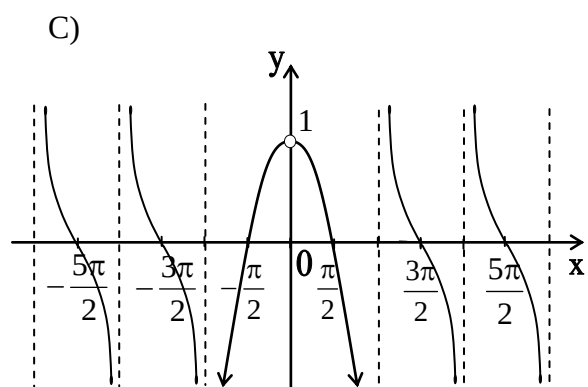
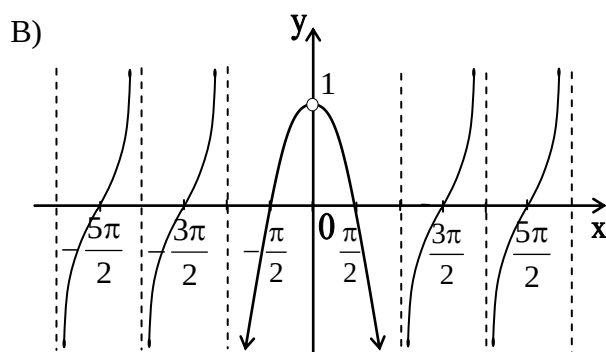
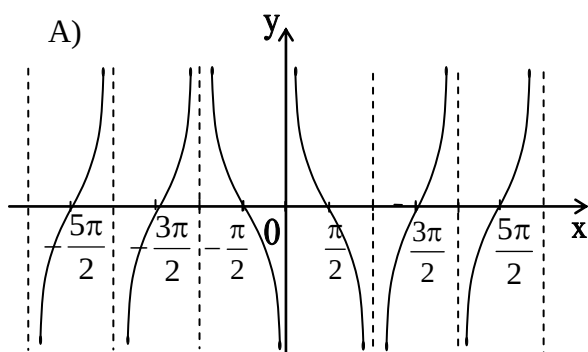
- A) ${}^{13}C_3 - 3^5 C_3 - 3$ B) ${}^{13}C_3 - 2^5 C_3 - 4$
 C) ${}^{13}C_3 - 3^5 C_3 - 4$ D) ${}^{13}C_3 - 3^5 C_3 - 5$
9. The base of a solid is the region bounded by the parabola $x = 4y - y^2$ and the y axis. Vertical cross sections are right angled isosceles triangles perpendicular to the x axis as shown.



Which integral represents the volume of this solid?

- A) $\int_0^4 2\sqrt{4-x} \, dx$ B) $\int_0^4 \pi(4-x) \, dx$ C) $\int_0^4 (8-2x) \, dx$ D) $\int_0^4 (16-4x) \, dx$

10. Which of the following shows the graph of $y = x \cot x$?



End of Section I

Section II

90 Marks

Attempt Questions 11-16

Allow about 2 hours and 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

All necessary working should be shown in every question.

Question 11 (15 marks) Use a SEPARATE writing booklet **Marks**

(a) The complex number z is given by $z = -\sqrt{3} + i$.

(i) Express z in modulus argument form. **2**

(ii) Hence show that $z^7 + 64z = 0$ **2**

(b) Find values A , B and C such that:

$$\frac{8(1-x)}{(2-x^2)(2-2x+x^2)} = \frac{A-Bx}{(2-2x+x^2)} - \frac{Cx}{(2-x^2)} \quad \textbf{3}$$

(c) Factorise $z^2 + 4iz + 5$ over the complex field. **1**

(d) Using the substitution $x = 2 \sin \theta$, show that **4**

$$\int_{-1}^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx = \pi - \sqrt{3}$$

(e) Sketch the region in the Argand diagram defined by $z\bar{z} + 2(z + \bar{z}) \leq 0$ **3**

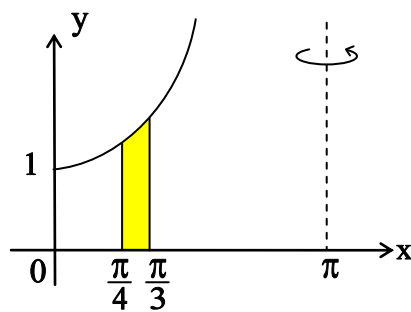
End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet

(a) Use the substitution $t = \tan \frac{x}{2}$ to evaluate $\int \frac{dx}{1 + \sin x + \cos x}$. **3**

(b) Consider the equation $z^3 + mz^2 + nz + 6 = 0$, where m and n are real. **3**
It is known that $1 - i$ is a root of the equation. Find the values of m and n .

(c) The area bounded by the curve $y = \sec^2 x$, the x -axis, **4**
 $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$ is rotated about the line $x = \pi$ to form a solid.



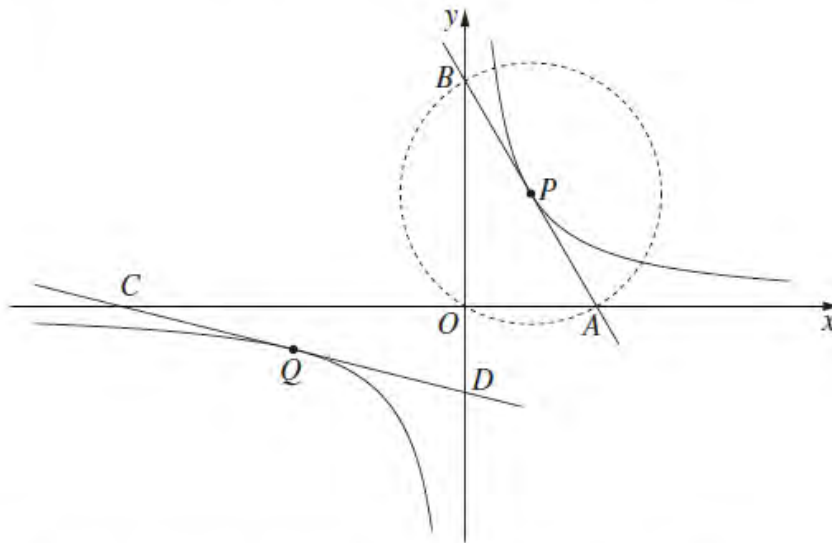
Use the method of cylindrical shells to find the volume of the solid.

Question 12 continues on page 8

Question 12 (continued)

- (d) The points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$, where $|p| \neq |q|$, lie on the rectangular hyperbola with equation $xy = c^2$.

The tangent to the hyperbola at P intersects the x -axis at A and the y -axis at B . Similarly, the tangent to the hyperbola at Q intersects the x -axis at C and the y -axis at D .



- | | |
|--|---|
| (i) Show that the equation of the tangent at P is $x + p^2y = 2cp$. | 2 |
| (ii) Show that A , B and O are on a circle with centre P . | 2 |
| (iii) Prove that BC is parallel to PQ . | 1 |

End of Question 12

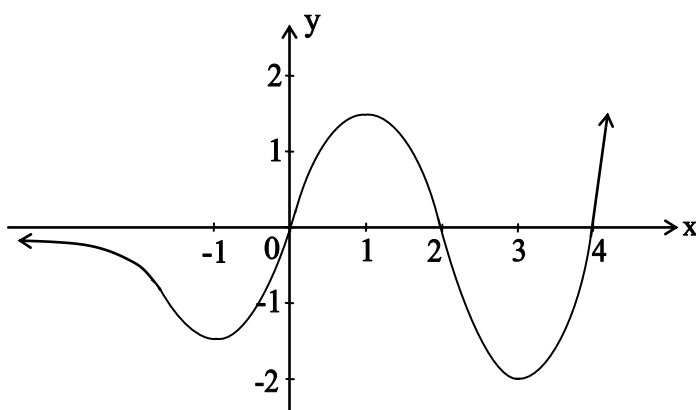
Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) (i) Let $I_n = \int_0^1 x^n \sqrt{1-x^3} dx$ for $n \geq 2$. Show that: 3

$$I_n = \frac{2n-4}{2n+5} \times I_{n-3} \text{ for } n \geq 5$$

- (ii) Hence find I_8 2

- (b) The graph of a certain function $y = f(x)$ is shown below.



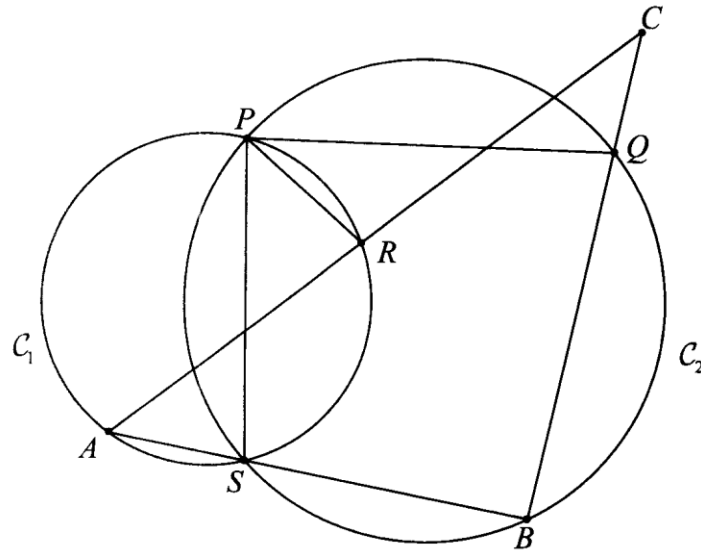
Sketch the following curves:

- (i) $y = \frac{1}{f(x)}$ 2
- (ii) $y = \ln [1 - f(x)]$ 2
- (iii) $y^2 = 1 + f(x)$ 2

Question 13 continues on page 10

Question 13 (continued)

- (c) Two circles C_1 and C_2 meet at P and S . Points A and R lie on C_1 and points B and Q lie on C_2 . AB passes through S and AR produced meets BQ produced at C , as shown in the diagram.

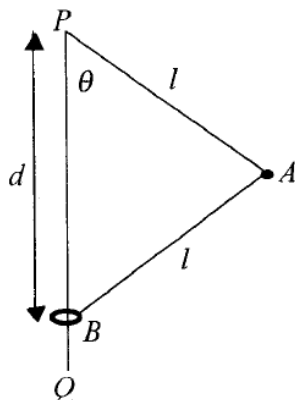


- | | |
|---|----------|
| (i) Prove that $\angle PRA = \angle PQB$. | 2 |
| (ii) Prove that the points P, R, Q and C are concyclic. | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

(a)



PQ is a smooth vertical rod. Particle A of mass m is attached to a point P by a string of length l and A is also attached by a second string of length l to a smooth ring B of mass M which is free to slide on the rod PQ without friction. A is set in motion in a horizontal circle about PQ with constant angular velocity ω . B is in equilibrium.

T_1 and T_2 are the tensions in the strings AP and AB respectively when AP makes an angle θ with the vertical.

- (i) Draw diagrams showing the forces acting on each of A and B . 1
- (ii) Hence show that $T_1 - T_2 = \frac{mg}{\cos \theta}$, $T_1 + T_2 = ml\omega^2$ and $T_2 = \frac{Mg}{\cos \theta}$. 2
- (iii) Deduce that $d = \frac{2g}{\omega^2} \left(1 + 2 \frac{M}{m} \right)$. 2

Question 14 continues on page 12

Question 14 (continued)

- (b) A particle's resistance to motion in a medium is proportional to mv^2 where m is the particle's mass and v is its velocity at time t .

- (i) Initially the particle is projected downwards in the medium where the speed of projection is equal to the terminal velocity V_T . 1

Show that $V_T^2 = \frac{g}{k}$ where k is the constant of proportionality.

The particle is now projected vertically upwards in the same medium.

- (ii) Show that $x = \frac{V_T^2}{2g} \ln \left(\frac{2V_T^2}{V_T^2 + v^2} \right)$. 2

- (iii) Hence show that $H = \frac{V_T^2 \ln 2}{2g}$ where H represents the particle's maximum height above its point of projection. 1

- (iv) Show that during the particle's ascent $v = V_T \tan \left(\frac{\pi}{4} - \frac{g}{V_T} t \right)$. 2

- (v) Hence show that $\frac{2V_T^2}{V_T^2 + v^2} = 1 + \sin \left(\frac{2g}{V_T} t \right)$. 2

- (vi) If T is the time taken to achieve *half* its maximum height, show that 2
 $T = \frac{V_T}{2g} \sin^{-1}(\sqrt{2} - 1)$.

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet

(a) (i) If $0 \leq a \leq b$ show that $\frac{a}{1+a} \leq \frac{b}{1+b}$ 2

(ii) Hence or otherwise show that $\frac{a}{1+a} \leq \frac{b}{1+b} + \frac{c}{1+c}$ where $a \leq b+c$ and $a, b, c \geq 0$ 2

(b) An urn contains 5 balls numbered from 1 to 5. A ball is chosen at random and its number is noted. The ball is then returned to the urn. This is done a total of five times.

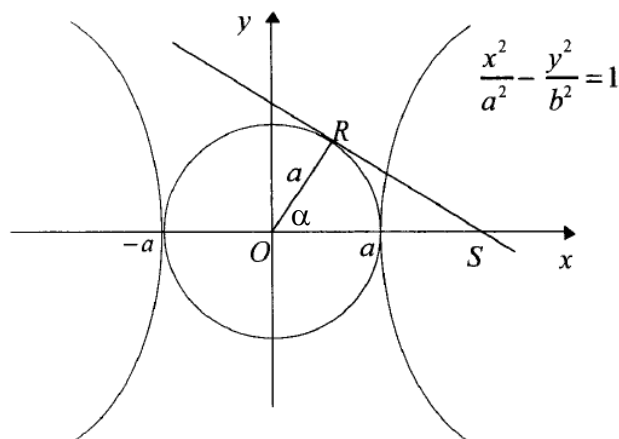
(i) What is the probability that each ball is selected exactly once? 1

(ii) What is the probability that at least one ball is not selected? 1

(iii) What is the probability of obtaining 1,1,2,3,4 in any order? 1

(iv) What is the probability that exactly one of the balls is not selected? 2

(c)



S is the focus of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, $a \neq b$, which lies on the positive x -axis.

R is a point on the auxiliary circle of the hyperbola such that R lies in the first quadrant and SR is tangent to the auxiliary circle. The eccentricity of the hyperbola is e and $\angle ROS = \alpha$.

(i) Show that R lies on a directrix of the hyperbola. 1

(ii) Show that SR has equation $y = -\frac{1}{\sqrt{e^2-1}}(x - ae)$. 1

(iii) If SR meets the hyperbola at the point $(a \sec \theta, b \tan \theta)$, show that $e^2(2 - e^2)\sec^2 \theta - 2e \sec \theta + \{e^2 + (e^2 - 1)^2\} = 0$. 2

(iv) By considering this as a quadratic equation in $\sec \theta$, deduce that SR intersects the hyperbola in two distinct points P and Q , lying on the same branch of the hyperbola if $e^2 < 2$ and lying on opposite branches if $e^2 > 2$. 2

End of Question 15

Question 16 (15 marks) Use a SEPARATE writing booklet

- (a) The n th Fermat number, F_n , is defined by $F_n = 2^{2^n} + 1$ for $n = 0, 1, 2, 3, \dots$, where 2^{2^n} means 2 raised to the power of 2^n . 4

Prove by mathematical induction, that for all positive integers:

$$F_0 \times F_1 \times F_2 \times \dots \times F_{n-1} = F_n - 2$$

- (b) A coin is tossed six times. What is the probability that there will be more tails on the first three of the six throws than on the last three throws? 3

- (c) Let $x = \cos \theta + i \sin \theta$ for $0 < \theta < 2\pi$, and let n be a positive integer.

- (i) Show that $x^k + \frac{1}{x^k} = 2 \cos k\theta$, for any positive integer k . 2

- (ii) Show that 3

$$\begin{aligned} \left(x + \frac{1}{x}\right)^{2n} &= \left(x^{2n} + \frac{1}{x^{2n}}\right) + \binom{2n}{1} \left(x^{2n-2} + \frac{1}{x^{2n-2}}\right) + \binom{2n}{2} \left(x^{2n-4} + \frac{1}{x^{2n-4}}\right) + \\ &\quad \dots + \binom{2n}{n-1} \left(x^2 + \frac{1}{x^2}\right) + \binom{2n}{n}. \end{aligned}$$

- (iii) Deduce that 1

$$\begin{aligned} 2^{2n-1} \cos^{2n} \theta &= \cos 2n\theta + \binom{2n}{1} \cos(2n-2)\theta + \binom{2n}{2} \cos(2n-4)\theta + \\ &\quad \dots + \binom{2n}{n-1} \cos 2\theta + \frac{1}{2} \binom{2n}{n}. \end{aligned}$$

- (iv) Hence show that $\int_0^{2\pi} \cos^{2n} \theta d\theta = \frac{\pi}{2^{2n-1}} \binom{2n}{n}$. 2

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$



2014 Year 12 Mathematics Extension 2 HSC Task 4 Trial HSC Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
1. D 2. B 3. A 4. B 5. D 6. A 7. C 8. C 9. C 10. D 1. D $25x^2 + 16y^2 = 400$ is equivalent to $\frac{x^2}{16} + \frac{y^2}{25} = 1$ which has its foci on the y axis. $e^2 = 1 - \frac{16}{25} = \frac{9}{25}$, so $e = \frac{3}{5}$ focal length is ae, $ae = 5 \times \frac{3}{5} = 3$ So foci are $(0, \pm 3)$ 2. B $\int \cot x \, dx = \int \frac{\cos x}{\sin x} \, dx$ $= \ln(\sin x) + c$ $= \ln\left(\frac{1}{\operatorname{cosec} x}\right) + c$ $= \ln 1 - \ln(\operatorname{cosec} x) + c$ $= -\ln(\operatorname{cosec} x) + c$ 3. A As $v = r\omega$ then $48\pi = 6\omega$ that is $\omega = 8\pi$ rad/s Now, the angular velocity is 8π and as each revolution is 2π radians then the particle makes 4 revolutions per second. 4. B $x^3 - 2x^2 + 3 = 0$ that is $x^3 - 2x^2 + 0x + 3 = 0$ This means $\alpha + \beta + \gamma = 2$ $\alpha\beta + \beta\gamma + \alpha\gamma = 0$ and $\alpha\beta\gamma = -3$ Now as α is a solution of the equation then $\alpha^3 - 2\alpha^2 + 3 = 0$, so $\alpha^3 = 2\alpha^2 - 3$ Similarly, $\beta^3 = 2\beta^2 - 3$ & $\gamma^3 = 2\gamma^2 - 3$ Therefore, $\alpha^3 + \beta^3 + \gamma^3 = 2(\alpha^2 + \beta^2 + \gamma^2) - 9$ But $(\alpha + \beta + \gamma)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(\alpha\beta + \beta\gamma + \alpha\gamma)$ So $(2)^2 = \alpha^2 + \beta^2 + \gamma^2 + 2(0)$ that is $\alpha^2 + \beta^2 + \gamma^2 = 4$ Hence, $\alpha^3 + \beta^3 + \gamma^3 = 2(4) - 9 = -1$		5. D $u = \ln x, \therefore du = \frac{dx}{x}$ $\int_1^e \frac{dx}{x\sqrt{1+\ln^2 x}}$ $= \int_0^1 \frac{du}{\sqrt{1+u^2}} = \left[\ln(u + \sqrt{1+u^2}) \right]_0^1$ $= \ln(1 + \sqrt{2}) - \ln 1$ $= \ln(1 + \sqrt{2})$ 6. A $\operatorname{Arg}\left(\frac{1}{1+i}\right) = \operatorname{Arg} 1 - \arg(1+i) = 0 - \frac{\pi}{4} = -\frac{\pi}{4}$ $\operatorname{Arg}\left(\frac{1}{(1+i)^2}\right) = \operatorname{Arg} 1 - \arg(1+i)^2 = 0 - 2 \times \frac{\pi}{4} = -\frac{\pi}{2}$ $\operatorname{Arg}\left(\frac{1}{(1+i)^{20}}\right) = \operatorname{Arg} 1 - \arg(1+i)^{20} = 0 - 20 \times \frac{\pi}{4} = -\frac{20\pi}{4}$ Let $S = \operatorname{Arg}\left(\frac{1}{1+i}\right) + \operatorname{Arg}\left(\frac{1}{(1+i)^2}\right) + \dots + \operatorname{Arg}\left(\frac{1}{(1+i)^{20}}\right)$ $= -\frac{\pi}{4} - \frac{2\pi}{4} - \frac{3\pi}{4} \dots \dots \dots - \frac{20\pi}{4}$ $= -\frac{\pi}{4}(1 + 2 + 3 + \dots \dots \dots + 20)$ (This is an Arithmetic Series with 20 terms, the first is $a = 1$ and the last being 20, so the sum is $\frac{20}{2}(1 + 20) = 210$) Therefore, $S = -\frac{\pi}{4}(210) = -52\frac{1}{2}\pi$. But an Arg is always between $-\pi$ and π Hence, $S = 26$ revolutions $-\frac{\pi}{2} = -\frac{\pi}{2}$	



2014 Year 12 Mathematics Extension 2 HSC Task 4 Trial HSC Solutions

7. C

$$\text{Arg}(az^2) = \text{Arg}(a) + 2\text{Arg}(z)$$

So option A is not correct as $\text{Arg}(az^2)$ in the diagram is incorrect.

$$\text{Mod}(az^2) = \text{mod}(a) \times (\text{mod}(z))^2$$

and $\text{mod}(a) = 1$ this means

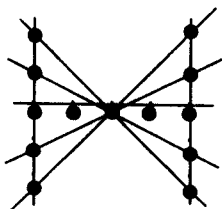
$$\text{Mod}(az^2) = (\text{mod}(z))^2 \text{ but } \text{Mod}(az^2) < 1$$

$$\text{Hence } (\text{mod}(z))^2 < 1$$

$$\text{and so } (\text{mod}(z))^2 < \text{mod}(z)$$

Only the diagram in option C satisfies this condition.

8. C



13 points can form combinations of 3 in ${}^{13}C_3$ ways.

But 5 points in a straight line cannot form a triangle, and there are 3 such lines. So $3 \times {}^5C_3$

must be subtracted from the total.

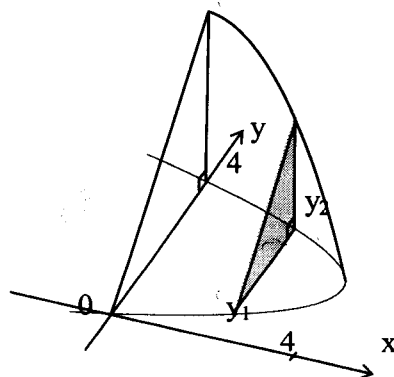
Also 3 points in a straight line cannot form a triangle, and there are 4 such lines.

So $4 \times {}^3C_3 = 4$ must be subtracted from the total,

Hence, the total number of triangles that can be formed is

$${}^{13}C_3 - 3 \times {}^5C_3 - 4.$$

9. C



The area of 1 cross section = $\frac{1}{2}(y_2 - y_1)^2$

Find an expression for y_1 and y_2 in terms of x .

$$x = 4y - y^2 \text{ so } y^2 - 4y + x = 0$$

$$\text{Hence } y = \frac{4 \pm \sqrt{16 - 4x}}{2}$$

$$= \frac{4 \pm 2\sqrt{4-x}}{2}$$

$$y = 2 \pm \sqrt{4-x}$$

$$\text{So } y_2 = 2 + \sqrt{4-x} \text{ and } y_1 = 2 - \sqrt{4-x}$$

$$\text{Hence } y_2 - y_1 = 2\sqrt{4-x}$$

$$\text{The area of a cross section} = \frac{1}{2}$$

$$(2\sqrt{4-x})^2 = 2(4-x)$$

$$\text{So the volume} = \int_0^4 (8-2x) dx$$

10. D

The graph of $y = x \cot x$ can be found using the

"multiplication of ordinates" method

Alternatively as $y = \cot x$ and $y = x$ are both

odd functions, $y = x \cot x$ must be even.

From the four options provided only D represents an even function.



2014 Year 12 Mathematics Extension 2 Task 4 Trial HSC SOLUTIONS

Suggested Solution (s)

Comments

Question 11

a) i) $2^7 \text{cis } \frac{5\pi}{6}$

ii) $z^7 + 64z$
 $= 2^7 \text{cis } \frac{35\pi}{6} + 64(2 \text{cis } \frac{5\pi}{6})$ (DMT)
 $= 128(\text{cis}(-\frac{\pi}{6}) + \text{cis}(\frac{5\pi}{6}))$
 $= 128(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} + \cos \frac{5\pi}{6} + i \sin \frac{5\pi}{6})$
 $= 128(\cos \frac{\pi}{6} - i \sin \frac{\pi}{6} - \cos \frac{\pi}{6} + i \sin \frac{\pi}{6})$
 $= 128 \times 0$
 $= 0$

as $\sin \theta = \sin(180 - \theta)$
 $\cos \theta = \cos(-\theta)$
 $\sin(-\theta) = -\sin \theta$
 $\cos(180 - \theta) = -\cos \theta$

(b) Let $8(1-x) \equiv (A-Bx)(2-x^2) - Cx(2-2x+x^2)$

when,

$x=0 \quad 8 \equiv 2A \quad \therefore A=4$ ✓

$x=1 \quad 0 = 4 - B - C$

$B+C=4$ — (1)

$x=-1 \quad 16 = 4 + B + 5C$

$B+5C=12$ — (2)

(1)-(2) $-4C=-8 \quad \therefore C=2, B=2$ ✓

$\therefore A=4, B=2, C=2$

(c) $z^2 + 4iz + 5$
 $= (z+5i)(z-i)$ ✓

$z \times -i$
 $z \times 5i$

(d) $x = 2 \sin \theta \quad x = -1, \theta = -\frac{\pi}{6}$
 $\frac{dx}{d\theta} = 2 \cos \theta \quad x = \sqrt{3}, \theta = \frac{\pi}{3}$ ✓

So $\int_{-1}^{\sqrt{3}} \frac{x^2}{\sqrt{4-x^2}} dx$

$= \int_{-\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{4 \sin^2 \theta}{\sqrt{4(1-\sin^2 \theta)}} \cdot 2 \cos \theta \cdot d\theta$

$= 4 \int_{-\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^2 \theta}{\cos \theta} \cdot \cos \theta d\theta$ ✓

$= 4 \int_{-\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{1}{2} - \frac{1}{2} \cos 2\theta d\theta$ ✓

$= 2 \left[\theta - \frac{1}{2} \sin 2\theta \right]_{-\frac{\pi}{6}}^{\frac{\pi}{3}}$ ✓ evaluation

$= 2 \left[\frac{\pi}{3} - \frac{1}{2} \sin \frac{2\pi}{3} - \left(-\frac{\pi}{6} - \frac{1}{2} \sin \left(-\frac{\pi}{3} \right) \right) \right]$

$= 2 \left[\frac{\pi}{2} - \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} \right]$

$= \pi - \sqrt{3}$

(e) $z\bar{z} + 2(z+z) \leq 0$

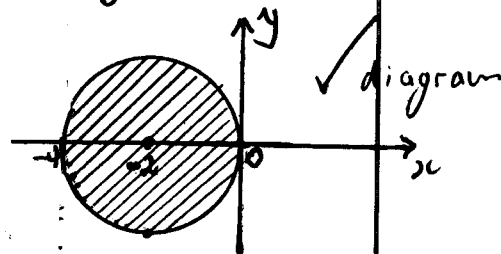
If $z = x+iy$

$(x+iy)(x-iy) + 2(x+iy+x-iy) \leq 0$

$x^2 + y^2 + 4x \leq 0$ ✓

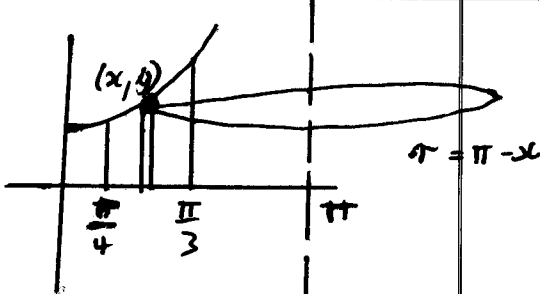
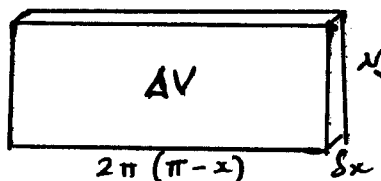
$x^2 + 4x + 4 + y^2 \leq 4$

$(x+2)^2 + y^2 \leq 4$ ✓





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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 12</u> (a) $t = \tan\left(\frac{x}{2}\right)$ $\therefore dx = \frac{2dt}{1+t^2}$ ✓ $\int \frac{dx}{1+\sin x + \cos x}$ $= \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2dt}{1+t^2}$ ✓ $= \int \frac{2}{1+t^2 + 2t + 1-t^2} dt$ $= \int \frac{2}{2(1+t)} dt$ $= \ln 1+t + C$ ✓ $= \ln 1+\tan\frac{x}{2} + C$ or (b) Since coeff's are real $1+i$ is also a root. so if $(1-i)(1+i)\alpha = -6$ product of roots $2\alpha = -6$ $\alpha = -3$ is third root Sum: $1-i + 1+i -3 = -m$ $-1 = -m$ $m = 1$ ✓ and $p(-3) = 0$ so $-27 + 9 - 3n + 6 = 0$ $3n = -12$ $n = -4$ ✓		(c)  Using method of cylindrical shells:  $\therefore \Delta V = 2\pi(\pi-x)y\delta x$ $V = 2\pi \int_{\pi/4}^{\pi/3} (\pi-x) \sec^2 x dx$ ✓ $= 2\pi^2 \int_{\pi/4}^{\pi/3} \sec^2 x dx - 2\pi \int_{\pi/4}^{\pi/3} x \sec^2 x dx$ ✓ $= 2\pi^2 \left[\tan x \right]_{\pi/4}^{\pi/3} - 2\pi \left[x \tan x \right]_{\pi/4}^{\pi/3} - \int_{\pi/4}^{\pi/3} \tan x dx$ $= 2\pi^2 (\sqrt{3} - 1) - 2\pi \left(\frac{\sqrt{3}\pi}{3} - \frac{\pi}{4} \right) - 2\pi \left[\ln(\cos x) \right]_{\pi/4}^{\pi/3}$ ✓ $= \frac{4\sqrt{3}\pi^2}{3} - \frac{3\pi^2}{2} + \pi \ln 2$ ✓ or equivalent	



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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
(d) i) $xy = c^2$ $\frac{d}{dx}(xy) = 0$ $\therefore x \cdot \frac{dy}{dx} + y = 0$ $\frac{dy}{dx} = -\frac{y}{x}$ at $x = cp$ $y = \frac{c}{p}$ $m_T = -\frac{c}{p} \times \frac{1}{cp}$ $= -\frac{1}{p^2}$ ✓ $\therefore y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$ ✓ $p^2y - cp = -x + cp$ $\therefore x + p^2y = 2cp$		iii) Similarly. $QD = QC$ Thus $\frac{AP}{AB} = \frac{DQ}{QC} = \frac{1}{2}$ $\therefore AD \parallel PQ \parallel BC$ ✓ (Intercepts on transversals cut by parallel lines are in the same ratio).	
ii) at A, $y = 0 \therefore x = 2cp$ $\therefore A(2cp, 0)$ at B, $x = 0 \therefore y = \frac{2c}{p}$ $\therefore B(0, \frac{2c}{p})$ Midpoint of AB is, $x = \frac{0 + 2cp}{2} = cp$ $y = \frac{\frac{2c}{p} + 0}{2} = \frac{c}{p}$ which are coordinates of P ✓ $\therefore PA = PB$ ✓ and since $\angle POB = 90^\circ$ ✓ A, B, O are on a circle centre P. (L in semi circle is 90°)			

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Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 13 (Continued) (c)(i) Let $\angle PRA = \alpha$ $\therefore \angle PSA = \alpha$ ($\angle$'s at circum. subtended by arc PA are equal) $\therefore \angle PSB = 180 - \alpha$ (Adjacent supplementary $\angle$'s on a straight line) $\therefore \angle PQB = \alpha$ (Opp. $\angle$'s of cyclic quad. are supplementary) $\therefore \angle PRA = \angle PQB$ ii) $\angle PRC = 180 - \alpha$ (Adjacent supplementary $\angle$'s on a straight line) $\angle PQC = 180 - \alpha$ (Adjacent supplementary $\angle$'s on a straight line) Since $\angle PRC = \angle PQC$ we have equal $\angle$s subtended by PC, which means P, R, Q and C are concyclic.	✓ ✓ ✓		



Suggested Solution (s)

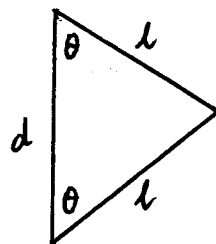
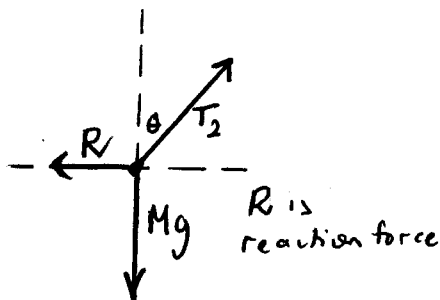
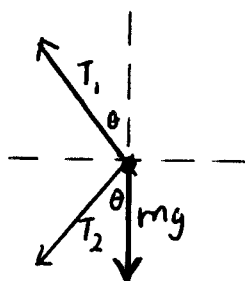
Comments

Question 14

a) i) Forces on A ✓

Forces on B

Dimensions Diagram



ii) Sum of Radial forces at A = $m\omega^2 r$

$$T_1 \sin \theta + T_2 \sin \theta = m\omega^2 r$$

$$T_1 + T_2 = \frac{m\omega^2 r}{\sin \theta} \quad \text{but } \sin \theta = \frac{r}{l}$$

$$T_1 + T_2 = m\omega^2 l \quad \dots (i) \quad \therefore l = \frac{r}{\sin \theta}$$

Vertical forces at A are balanced

$$T_1 \cos \theta = T_2 \cos \theta + mg$$

$$T_1 - T_2 = \frac{mg}{\cos \theta} \quad \dots (ii)$$

Vertical Forces at B are balanced

$$T_2 \cos \theta = Mg \quad \dots (iii)$$

$$T_2 = \frac{Mg}{\cos \theta}$$

iii) Subtracted (i) and (ii) $2T_2 = m\omega^2 l - \frac{mg}{\cos \theta}$

$$2\left(\frac{Mg}{\cos \theta}\right) = m\omega^2 l - \frac{mg}{\cos \theta}$$

$$2Mg = m\omega^2 l \cos \theta - mg$$

$$m\omega^2 l = \frac{g(m + 2M)}{\cos \theta}$$

$$l \cos \theta = \frac{g}{\omega^2} \left(1 + \frac{2M}{m}\right)$$

but $\cos \theta = \frac{d}{l} \quad l \cos \theta = \frac{d}{2}$

$$\therefore d = \frac{2g}{\omega^2} \left(1 + \frac{2M}{m}\right)$$



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Suggested Solution (s)	Comments
(b) (i) Downwards $\downarrow + \quad \downarrow mg \quad \uparrow mkv^2$ $\therefore m\ddot{x} = mg - mkv^2$ is resultant force $\ddot{x} = g - kv^2$ At terminal velocity $\ddot{x} = 0$. $\therefore kV_T^2 = g$ $V_T^2 = \frac{g}{k}$ ii) Upwards $\uparrow + \quad \downarrow mg \quad \downarrow mkv^2$ $\therefore m\ddot{x} = -mg - mkv^2$ is resultant force $\therefore v \frac{dv}{dx} = -(g + kv^2)$ $\frac{dx}{dv} = \frac{-v}{g + kv^2}$ $x = -\frac{1}{2k} \int \frac{2kv}{g + kv^2} dv$ $\therefore x = -\frac{1}{2k} \ln(g + kv^2) + C$ at $x=0, v=V_T \quad \therefore C = \frac{1}{k} \ln(g + kV_T^2)$ So $x = \frac{1}{2k} \ln \frac{g + kV_T^2}{g + kv^2}$ $= \frac{1}{2k} \ln \frac{kV_T^2 + kV_T^2}{kV_T^2 + kv^2}$ $= \frac{V_T^2}{2g} \ln \left(\frac{2V_T^2}{V_T^2 + v^2} \right)$ $g = kV_T^2$ ad $\frac{1}{k} = \frac{V_T^2}{g}$	



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Suggested Solution (s)	Comments
iii) At maximum height $x = H$, $v = 0$ $\therefore H = \frac{V_T^2}{2g} \ln\left(\frac{2V_T^2}{V_T^2 + 0}\right) \checkmark$ $= \frac{V_T^2}{2g} \ln(2V_T^2)$ iv) Starting with $\ddot{x} = -(g + kv^2)$ $\frac{dv}{dt} = \frac{-1}{k\left(\frac{g}{k} + v^2\right)}$ $\therefore t = -\frac{1}{k} \int \frac{1}{V_T^2 + v^2} dv$ $= -\frac{1}{k} \cdot \frac{1}{V_T} \tan^{-1}\left(\frac{v}{V_T}\right) + C$ at $t=0$ $v=V_T$ $\therefore C = \frac{1}{kV_T} \tan^{-1}(1) = \frac{\pi}{4kV_T}$ So $t = \frac{1}{kV_T} \left(\frac{\pi}{4} - \tan^{-1}\left(\frac{v}{V_T}\right)\right)$ $\therefore \tan^{-1}\left(\frac{v}{V_T}\right) = \frac{\pi}{4} - kV_T t$ $v = V_T \tan\left(\frac{\pi}{4} - kV_T t\right) \text{ but } kV_T = \frac{g}{V_T}$ $\therefore v = V_T \tan\left(\frac{\pi}{4} - \frac{g}{V_T} t\right) \text{ as required}$	



Suggested Solution (s)

Comments

(v) Algebraically and using trig identities

$$\frac{v}{V_T} = \tan\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)$$

$$\frac{v^2}{V_T^2} = \tan^2\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)$$

so $1 + \frac{v^2}{V_T^2} = 1 + \tan^2\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)$

$\therefore 1 + \frac{v^2}{V_T^2} = \sec^2\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)$ ✓

$$\frac{V_T^2 + v^2}{V_T^2} = \sec^2\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)$$

reciprocals

$$\frac{V_T^2}{V_T^2 + v^2} = \cos^2\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)$$

$$= \frac{1}{2} + \frac{1}{2} \cos\left(2\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)\right)$$

$$\therefore \frac{2V_T^2}{V_T^2 + v^2} = 1 + \cos\left(\frac{\pi}{2} - \frac{2gt}{V_T}\right)$$

$$= 1 + \sin\left(\frac{2gt}{V_T}\right) \text{ as required.}$$

Alternatively,

$$x = \int V_T \tan\left(\frac{\pi}{4} - \frac{g}{V_T} t\right) dt$$

$$= V_T \int \frac{\sin\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)}{\cos\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)} dt$$

$$= V_T \times \frac{-V_T}{g} \times \int \frac{-\sin\left(\frac{\pi}{4} - \frac{g}{V_T} t\right) \cdot \frac{-g}{V_T}}{\cos\left(\frac{\pi}{4} - \frac{g}{V_T} t\right)} dt$$



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Suggested Solution (s)	Comments
$\therefore x = \frac{V_T^2}{g} \ln \left[\cos \left(\frac{\pi}{4} - \frac{g}{V_T} t \right) \right] + C$ at $t=0, C=0 \therefore C = -\frac{V_T^2}{g} \ln \left(\frac{1}{\sqrt{2}} \right)$ $\therefore x = \frac{V_T^2}{g} \ln \left[\sqrt{2} \cos \left(\frac{\pi}{4} - \frac{g}{V_T} t \right) \right] \checkmark$ Equating expressions for displacement $\frac{V_T^2}{2g} \ln \left[\frac{2V_T^2}{V_T^2 + v^2} \right] = \frac{V_T^2}{g} \ln \left[\sqrt{2} \cos \left(\frac{\pi}{4} - \frac{g}{V_T} t \right) \right]$ log laws: $\ln \left[\frac{2V_T^2}{V_T^2 + v^2} \right]^{\frac{1}{2}} = \ln \left[\sqrt{2} \cos \left(\frac{\pi}{4} - \frac{g}{V_T} t \right) \right]$ equating subjects and squaring $\frac{2V_T^2}{V_T^2 + v^2} = 2 \cos^2 \left(\frac{\pi}{4} - \frac{g}{V_T} t \right)$ $= 2 \left(\frac{1}{2} + \frac{1}{2} \cos \left[2 \left(\frac{\pi}{4} - \frac{g}{V_T} t \right) \right] \right)$ $= 1 + \cos \left(\frac{\pi}{2} - \frac{2g}{V_T} t \right)$ $= 1 + \sin \frac{2g}{V_T} t \text{ as required}$ vi) Noting $\frac{1}{2}H = \frac{V_T^2 \ln 2}{4g}$ $\checkmark$ $\therefore \frac{V_T^2 \ln 2}{4g} = \frac{V_T^2}{2g} \ln \left(\frac{2V_T^2}{V_T^2 + v^2} \right)$ from part i) $\therefore \frac{1}{2} \ln 2 = \ln \left(\frac{2V_T^2}{V_T^2 + v^2} \right)$ $\therefore \frac{2V_T^2}{V_T^2 + v^2} = 2^{\frac{1}{2}} = 1 + \sin \left(\frac{2g}{V_T} t \right)$ from part v) rearranging gives $T = \frac{V_T}{2g} \sin^{-1}(\sqrt{2}-1)$	$\checkmark$ correct process



Suggested Solution (s)

Comments

Question 15

(a) i) Since $a \leq b$

$$a + ab \leq b + ab \quad \checkmark$$

$$a(1+b) \leq b(1+a)$$

$$\therefore \frac{a}{1+a} \leq \frac{b}{1+b} \quad \checkmark$$

ii) Replacing b with $b+c$ from the inequality in part i)

$$\frac{a}{1+a} \leq \frac{b+c}{1+b+c} \quad \checkmark$$

$$\leq \frac{b}{1+b+c} + \frac{c}{1+b+c}$$

$$\leq \frac{b}{1+b} + \frac{c}{1+c}$$

} progress $\checkmark$

(b) i) $P(E) = \frac{5}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{2}{5} \times \frac{1}{5} = \frac{5!}{5^5} = \frac{24}{625} \quad \checkmark$

ii) $P(E) = 1 - \frac{5!}{5^5} = \frac{601}{625} \quad \checkmark$

iii) $\frac{5!}{2!} / 5^5 = \frac{12}{625} \quad \checkmark$

iv) Given that there can be also double 2's, 3's or 4's $\checkmark$
When one ball is not selected, there are $4 \times 5!$ ways of leaving out the five. As each of the other $2!$ numbers can be left out, there are $5 \times 4 \times 5!$ permutations of exactly one ball not being selected. $\therefore P(E) = \frac{(20 \times 5!)}{2!} \div 5^5 = \frac{48}{125}$

(c) i) Since $S(ae, 0)$ $\sec \alpha = \frac{OS}{OR} = \frac{ae}{a} = e$ $\checkmark$ At R , $x = a \cos \alpha = \frac{a}{e}$

ii) $m_{OR} = \tan \alpha = \sqrt{\sec^2 \alpha - 1} = \sqrt{e^2 - 1} \quad \checkmark$ $\therefore R$ lies on directorix $\therefore m_{RS} = \frac{-1}{\sqrt{e^2 - 1}}$
 $\therefore SR$ is $y = \frac{-1}{\sqrt{e^2 - 1}}(x - ae)$

iii) $x = a \sec \theta$, $y = b \tan \theta$ $\therefore b \sqrt{e^2 - 1} = -(a \sec \theta - ae)$
Squaring: $b^2(e^2 - 1) \tan^2 \theta = a^2(\sec^2 \theta - 2e \sec \theta + e^2)$ and since $\frac{b^2}{a^2} = (e^2 - 1)$ for hyperbola
 $(e^2 - 1)^2(\sec^2 \theta - 1) = \sec^2 \theta - 2e \sec \theta + e^2$ $\checkmark$ Rearranging leads to quadratic.

iv) If SR is not a tangent to the hyperbola, the quadratic above will have two distinct real roots. $\checkmark$
Product of Roots = $\frac{e^2 + (e^2 - 1)^2}{e^2(2 - e^2)}$ So roots have same sign if $(2 - e^2) > 0$ i.e. $e^2 < 2$ $\checkmark$
and opposite sign if $(2 - e^2) < 0$ i.e. $e^2 > 2$ $\checkmark$
 $\therefore$ Same branch as P if $e^2 < 2$
 $\therefore$ opposite branch to P if $e^2 > 2$



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Suggested Solution (s)

Comments

Question 16

(a) When $n=1$, L.H.S = F_0 R.H.S = $F_1 - 2$
 $= 2^{(2^0)} + 1$ $= 2^{(2^1)} + 1 - 2$ ✓
 $= 3$ $= 3$

∴ statement is true when $n=1$.

Assume the statement is true for $n=k$, some fixed positive integer

ie. $F_0 \times F_1 \times F_2 \times \dots \times F_{k-1} = F_k - 2$

When $n=k+1$, L.H.S = $F_0 \times F_1 \times F_2 \times \dots \times F_{n-1}$
 $= F_0 \times F_1 \times F_2 \times \dots \times F_{k-1} \times F_k$
 $= (F_k - 2) \times F_k$ by assumption ✓
 $= (F_k)^2 - 2F_k$
 $= (2^{2^k} + 1)^2 - 2(2^{2^k} + 1)$
 $= 2^{2 \times 2^k} + 2 \times 2^{2^k} + 1 - 2 \times 2^{2^k} - 2$ ✓
 $= 2^{2^{k+1}} + 1 - 2$
 $= (2^{2^n} + 1) - 2$
 $= F_n - 2$ as required ✓

If statement is true for $n=k$, it has been proved true for $n=k+1$,
 Since true for $n=1$, then proved true for $n=2, 3, 4, \dots$

(b) Considering the first three and last three throws separately,
 the are: Equal tails, More Tails, Less Tails ✓

Taking $P(\text{Equal Tails}) = P(1H) + P(2H) + P(3H) + P(0H)$
 $= 9\left(\frac{1}{2}\right)^6 + 9\left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6 + \left(\frac{1}{2}\right)^6$
 $= \frac{20}{64}$ ✓

∴ $P(\text{More tails in first 3 throws}) = \frac{1}{2} \times \left(1 - \frac{20}{64}\right)$
 $= \frac{11}{32}$ ✓



(c) (i)

By DeMoivre's theorem,

$$x^k = (\cos \theta + i \sin \theta)^k = \cos k\theta + i \sin k\theta$$

$$x^{-k} = (\cos \theta + i \sin \theta)^{-k} = \cos(-k\theta) + i \sin(-k\theta) = \cos k\theta - i \sin k\theta \quad \checkmark$$

$$x^k + x^{-k} = \cos k\theta + i \sin k\theta + \cos k\theta - i \sin k\theta \quad \checkmark$$

$$= 2 \cos k\theta$$

(ii)

$$\left(x + \frac{1}{x}\right)^{2n} = \binom{2n}{0} x^{2n} + \binom{2n}{1} x^{2n-2} + \binom{2n}{2} x^{2n-4} + \dots + \binom{2n}{n} + \dots + \binom{2n}{2n-2} \frac{1}{x^{2n-2}} + \binom{2n}{2n-1} \frac{1}{x^{2n-1}} + \binom{2n}{2n} \frac{1}{x^{2n}}$$

$$= \binom{2n}{0} x^{2n} + \binom{2n}{1} x^{2n-2} + \binom{2n}{2} x^{2n-4} + \dots + \binom{2n}{n} + \dots + \binom{2n}{2} \frac{1}{x^{2n-4}} + \binom{2n}{1} \frac{1}{x^{2n-2}} + \binom{2n}{0} \frac{1}{x^{2n}} \quad \checkmark \text{ expansion}$$

$$\text{since } \binom{2n}{2n-k} = \binom{2n}{k} \quad \checkmark \text{ of identity.}$$

$$\left(x + \frac{1}{x}\right)^{2n} = \binom{2n}{0} \left(x^{2n} + \frac{1}{x^{2n}}\right) + \binom{2n}{1} \left(x^{2n-2} + \frac{1}{x^{2n-2}}\right) + \binom{2n}{2} \left(x^{2n-4} + \frac{1}{x^{2n-4}}\right) + \dots + \binom{2n}{n-1} \left(x^2 + \frac{1}{x^2}\right) + \binom{2n}{n}$$

$$\text{Note } \binom{2n}{0} = 1$$

(iii)

Using the result from part i, $x^k + \frac{1}{x^k} = 2 \cos k\theta$ in the identity from part ii:

$$(2 \cos \theta)^{2n} = \binom{2n}{0} (2 \cos 2n\theta) + \binom{2n}{1} (2 \cos(2n-2)\theta) + \binom{2n}{2} (2 \cos(2n-4)\theta) + \dots + \binom{2n}{n-1} (2 \cos 2\theta) + \binom{2n}{n} \quad \checkmark$$

Dividing both sides by 2:

$$2^{2n-1} \cos^{2n} \theta = \cos 2n\theta + \binom{2n}{1} \cos(2n-2)\theta + \binom{2n}{2} \cos(2n-4)\theta + \dots + \binom{2n}{n-1} \cos 2\theta + \frac{1}{2} \binom{2n}{n}$$



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(iv)

$$\int_0^{2\pi} 2^{2n-1} \cos^{2n} \theta d\theta = \int_0^{2\pi} \cos 2n\theta + \binom{2n}{1} \cos(2n-2)\theta + \binom{2n}{2} \cos(2n-4)\theta + \dots + \binom{2n}{n-1} \cos 2\theta + \frac{1}{2} \binom{2n}{n} d\theta$$

Since $\int_0^{2\pi} \cos k\theta d\theta = 0$ for all even integers k , all the integrals on the right hand side are zero except for the constant term.

$$\begin{aligned} \int_0^{2\pi} 2^{2n-1} \cos^{2n} \theta d\theta &= \int_0^{2\pi} \frac{1}{2} \binom{2n}{n} d\theta \\ &= \left[\frac{1}{2} \binom{2n}{n} \theta \right]_0^{2\pi} \\ &= \pi \binom{2n}{n} \end{aligned}$$

Dividing both sides by 2^{2n-1} ,

$$\int_0^{2\pi} \cos^{2n} \theta d\theta = \frac{\pi}{2^{2n-1}} \binom{2n}{n}.$$